Data Analytics Capstone

Predicting Medicare Part D Drug Spending Patterns and Identifying Prescriber Outliers: A Data-Driven Framework for Pharmacy Benefit Management Formulary Optimization

Interim Report

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QM640: Data Analytics Capstone

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GitHub Repository: https://github.com/Apena87/QM-640-Quantitative-Methods-Capstone

Introduction

Background and Context

Prescription drug spending is one of the most significant and fastest-growing cost drivers in U.S. healthcare. The Medicare Part D program, which provides outpatient drug coverage to approximately 50.8 million beneficiaries, reported gross drug costs of $275.8 billion in 2023, a 166% increase from $103.7 billion in 2013, representing a compound annual growth rate of approximately 10.3% (Centers for Medicare & Medicaid Services [CMS], 2023a). Net program expenditures reached $101.9 billion in 2022, making Part D one of the largest single components of federal healthcare spending (Medicare Payment Advisory Commission [MedPAC], 2024). The scale and trajectory of these expenditures have made Part D a sustained focus of federal cost-containment policy, especially in the wake of the Inflation Reduction Act of 2022 and it's direct drug price negotiation provisions.

The cost burden within Part D is disproportionately concentrated among brand-name drugs. Although brand-name drugs comprise only 14.9% of total Part D claim volume, they account for 85.2% of total gross drug costs, up from 75.0% in 2013. By 2023, the average brand-name claim cost $976, which is approximately 33 times the average generic claim cost of $29.25. This disparity is compounded by the growth of specialty drugs, which include high-cost biologic agents for oncology, immunology, and rare diseases. Niu et al. (2024) documented that specialty drug spending as a share of total Part D gross costs grew substantially over the 2012 to 2021 period, with oncology and immunology representing the largest absolute dollar contributors. These trends have elevated formulary design and utilization management as primary levers for pharmacy benefit managers (PBMs) and health plans seeking to moderate expenditure growth.

Prescriber behavior is a key intermediate variable connecting drug availability and patient utilization. Multiple lines of research establish that financial relationships between pharmaceutical manufacturers and physicians as documented in the federal Open Payments database are associated with higher-cost brand-name prescribing. Mitchell et al. (2021) conducted a systematic review of 36 studies and found that 30 reported a positive association between pharmaceutical payments and prescribing, with 25 demonstrating a dose-response relationship. Sharma et al. (2018) linked Open Payments and Part D data for 667,278 physicians and found odds ratios of 2.21 to 8.06 for brand-name prescribing across six drug categories among payment recipients. These findings motivate a national-scale analysis of the relationship between prescriber financial relationships and formulary cost efficiency within Medicare Part D.

Data-driven approaches to understanding prescribing variation and drug cost trends have become increasingly feasible as CMS has expanded its public data infrastructure. The Part D Prescribers by Provider and Drug file contains over 1.1 million records linking individual National Provider Identifiers (NPIs) to drug-level claim volumes and costs, enabling prescriber-level analysis at national scale. Rosenberg et al. (2018) demonstrated the utility of K-means clustering and visualization applied to this dataset, identifying substantial variation in prescribing patterns within specialties and across geographic regions. The Congressional Budget Office (2022) and the Assistant Secretary for Planning and Evaluation (ASPE, 2022) have independently confirmed that specialty drug uptake and price growth, rather than volume expansion, are the primary drivers of per-capita spending increases. This project builds on that analytical foundation to develop actionable, quantitative insights for PBM formulary decision-making.

Problem Statement

Despite the availability of rich public CMS datasets, it was my experience working in the PBM space that health plan administrators lack an integrated, data-driven framework that simultaneously identifies the structural predictors of per-drug Part D spending growth and characterizes the distinct segments of the drug formulary by cost profile. Current formulary management largely relies on manufacturer-negotiated rebates and therapeutic equivalence reviews, without systematically incorporating prescriber-level variation data or unsupervised pattern recognition to prioritize intervention targets. This study addresses that gap by applying multiple linear regression and K-means clustering to publicly available CMS administrative data to identify which drugs, drug attributes, and prescriber characteristics are most strongly associated with elevated Part D expenditure.

Purpose of the Study

This study aims to accomplish five analytical objectives. First, it identifies the drug-level attributes that are statistically significant predictors of total Part D spending in 2023, using an OLS multiple linear regression with log-transformed outcomes. Second, it characterizes state-level variation in brand-versus-generic prescribing rates and tests which demographic and plan factors predict higher brand utilization. Third, it examines whether the presence of Open Payments financial relationships is independently associated with higher-cost prescribing in drug classes where lower-cost generic alternatives exist. Fourth, it applies K-means clustering to identify distinct drug utilization profiles within the Part D formulary that differ significantly in per-beneficiary cost and spending growth trajectory. Fifth, it constructs a scenario simulation estimating the annual savings opportunity if high-cost prescribers shifted their brand-to-generic ratios to match median prescriber behavior within their specialty. Together, these objectives

produce a prioritized list of formulary and prescriber engagement interventions estimated by potential cost impact.

Interim Project Status (Progress Snapshot)

The following summarizes project completion status as of this interim submission:

- **Completed:** Data collection (all three CMS files downloaded and validated); data cleaning and parsing of 3,598 drug-level records and 200,000-record prescriber sample; EDA including 11 figures covering spending trends, brand/generic dynamics, unit cost distribution, and geographic variation; OLS regression (RQ1) with feature engineering; K-means clustering (RQ4) with elbow method optimization and cluster profiling.
- **In Progress:** Full prescriber file processing (expanding 200K sample to 1.1M records); ANOVA for state-level brand/generic variation (RQ2); Open Payments database acquisition and NPI linkage to the prescriber file (RQ3).
- **Pending:** Open Payments prescribing correlation analysis (RQ3); brand-to-generic savings simulation model (RQ5); cross-RQ findings integration; final report narrative and recommendations.

Scope and Objectives

This project examines Medicare Part D using publicly available CMS administrative data spanning calendar years 2013 through 2023, with three primary datasets: the CMS Part D Drug Spending by Drug file (2019-2023), the Part D Prescribers by Provider and Drug file (2023), and the Part D Grand Totals by Calendar Year file (2013-2023). All data are freely downloadable from CMS with no paywall or registration requirement. The overarching research problem is: What structural factors drive per-beneficiary Medicare Part D drug spending, and how can quantitative models support payer and pharmacy benefit manager (PBM) decision-making around formulary design, utilization management, and prescriber engagement? The five research questions below operationalize this problem across distinct analytical methods, as summarized in Table 1.

Table 1 *Research Questions and Corresponding Analytical Methods*

RQ	Research Question	Method
RQ1	What drug attributes are statistically significant predictors of total spending growth from 2019 to 2023?	OLS Multiple Linear Regression; Trend Analysis
RQ2	Are there significant state-level variations in brand-versus-generic prescribing rates, and what demographic factors predict higher brand utilization?	OLS Regression; ANOVA
RQ3	Does the presence of Open Payments financial relationships correlate with higher-cost prescribing in classes with generic alternatives?	Pearson Correlation; Binary Logistic Regression
RQ4	Can unsupervised clustering identify distinct drug utilization profiles that differ significantly in mean per-beneficiary drug cost?	K-Means Clustering; ANOVA Post-Hoc Test
RQ5	What is the estimated annual savings opportunity if high-cost prescribers shifted brand-to-generic ratios to match median prescriber behavior within specialty?	Scenario Simulation (Monte Carlo)

Note. RQ = Research Question. OLS = Ordinary Least Squares. All analyses use publicly available CMS data. RQ1 and RQ4 are complete at interim; RQ2, RQ3, and RQ5 are in progress or pending.

Literature Survey

Literature Review Approach

Sources were identified through three complementary strategies. First, PubMed and Google Scholar searches were conducted using the keyword combinations: Medicare Part D AND (spending OR expenditure), pharmaceutical industry payments AND prescribing behavior, generic substitution AND Medicare, K-means clustering AND prescribers, and pharmacy benefit management AND formulary optimization. Second, federal institutional sources -- CMS, MedPAC, CBO, ASPE, and the Dartmouth Institute were searched directly for relevant program reports and data releases. Third, references cited in high-yield papers (Mitchell et al., 2021; Rosenberg et al., 2018) were reviewed for additional relevant literature. Inclusion criteria required each source to address at least one of the five research questions, use empirical methods, and be published or issued by a credible institution. All 14 sources retained inform specific methodological or contextual decisions in this study, as detailed in Table 2.

Summary of Key Literature (Minimum 10 Relevant Sources)

Table 2 *Literature Relevance Matrix*

Author (Year)	Domain/Context	Dataset/Setting	Method(s)	Key Findings	How it supports RQ(s)
MedPAC (2024)	Medicare Part D program finance	Federal program-level data	Descriptive analysis	Net Part D expenditures reached $101.9B in 2022; specialty drug costs accelerating	Anchors program-level spending figures; benchmarks for RQ1 and RQ2
Niu et al. (2024)	Specialty drug cost	CMS Part D 2012-2021	Longitudinal spending	Specialty drugs (oncology,	Establishes specialty

Author (Year)	Domain/Context	Dataset/Setting	Method(s)	Key Findings	How it supports RQ(s)
	concentration		analysis	immunology) account for growing share of gross Part D costs	penetration as primary cost driver; informs RQ1 cluster interpretation
Mitchell et al. (2021)	Pharma payments and prescribing	36 studies systematic review	Systematic review and meta-analysis	30/36 studies found positive association; 25/36 found dose-response relationship	Primary empirical support for RQ3 Open Payments analysis; justifies hypothesis
Sharma et al. (2018)	Industry payments and brand prescribing	Open Payments + Part D, n=667,278 physicians	Logistic regression	OR 2.21-8.06 higher odds of brand prescribing for payment recipients across 6 drug classes	Provides OR benchmarks for RQ3; informs variable selection for prescriber model
Rosenberg et al. (2018)	Prescribing variation visualization	CMS Part D Prescribers file	K-means clustering; t-SNE	Substantial within-specialty and geographic prescribing variation identified	Directly informs K-means approach for RQ4; validates use of prescriber file
Buttorff et al. (2020)	Generic dispensing variation	CMS Part D 10-year panel	Regression; variance analysis	Persistent state-level generic rate variation attributed to plan type, premiums, state factors	Provides validated predictor candidates for state-level RQ2 model
Yeh et al. (2016)	Payments and statin prescribing	Massachusetts physicians + Part D	Cross-sectional regression	Payment recipients more likely to prescribe brand statins; dose-response confirmed	Proof-of-concept motivating national-scale Open Payments analysis in RQ3
CBO (2022)	U.S. drug expenditure trends	National spending accounts	Descriptive policy analysis	Gross Part D spending exceeded $200B in 2021; price growth primary driver	Corroborates CMS spending magnitudes; supports RQ1 contextual

Author (Year)	Domain/Context	Dataset/Setting	Method(s)	Key Findings	How it supports RQ(s)
					framing
ASPE (2022)	Per-capita drug spending	U.S. national (2016-2021)	Trend analysis	Per-capita spending rose substantially due to specialty and price, not volume	Confirms Part D reflects national trends; supports background for RQ1
Dartmouth Institute (n.d.)	Regional prescribing variation	CMS Part D regional data	Geographic analysis	Per-beneficiary spending varies >2.5x across hospital referral regions ($1,770-$4,738)	Benchmark for RQ2 geographic variation; motivates RQ5 savings simulation

Note. OR = odds ratio. MedPAC = Medicare Payment Advisory Commission. CBO = Congressional Budget Office. ASPE = Assistant Secretary for Planning and Evaluation. Full citations are provided in the Bibliography.

The literature converges on three themes directly relevant to this project. The first theme is cost concentration and specialty drug growth. MedPAC (2024), Niu et al. (2024), and CBO (2022) consistently document that a small subset of drugs like high-cost specialty biologics accounts for a disproportionate and growing share of Part D expenditures. This finding motivates the K-means clustering approach (RQ4), which aims to isolate that high-impact segment and characterize its distinguishing attributes. The Dartmouth Institute geographic data further confirms that this concentration is not evenly distributed geographically, underscoring the relevance of state-level modeling in RQ2.

The second theme is prescriber financial relationships and brand prescribing behavior. Three independent studies performed by Mitchell et al. (2021), Sharma et al. (2018), and Yeh et al. (2016) report consistent, statistically significant positive associations between pharmaceutical industry payments and brand-name prescribing. The systematic review by Mitchell et al. (2021) is particularly important as it synthesizes 36 peer-reviewed studies and identifies dose-response

relationships, substantially strengthening the causal inference. These findings provide the primary justification for RQ3 and for including Open Payments linkage in the final analysis.

The third theme is analytical methodology and precedent for data-driven formulary analysis. Rosenberg et al. (2018) demonstrate that K-means clustering applied to the CMS prescriber file successfully identifies meaningful prescribing subgroups, validating the methodological approach used in this study. Buttorff et al. (2020) show that state-level generic dispensing variation is large, stable over time, and attributable to identifiable plan and demographic factors -- directly informing the predictor selection for the RQ2 regression. ASPE (2022) and CBO (2022) confirm that price dynamics, not volume, drive per-capita cost growth, supporting the decision to include unit cost as a primary predictor in the OLS model.

Data Description

This project uses three publicly available CMS datasets. All are freely downloadable at data.cms.gov with no registration or paywall requirements. Combined, the three files provide complete coverage of Medicare Part D from 2013 through 2023 at the program, drug, and prescriber levels of analysis. A fourth supplementary dataset from the CDC Behavioral Risk Factor Surveillance System (BRFSS) is included as a source of state-level demographic controls for RQ2.

Data Source(s) and Access

Centers for Medicare & Medicaid Services. (2023a). Medicare Part D prescribers -- grand totals by calendar year [Data set]. https://data.cms.gov/provider-summary-by-type-of-service/medicare-part-d-prescribers

Centers for Medicare & Medicaid Services. (2023b). Medicare Part D spending by drug [Data set]. https://data.cms.gov/summary-statistics-on-use-and-payments/medicare-medicaid-spending-by-drug/medicare-part-d-spending-by-drug

Centers for Medicare & Medicaid Services. (2023c). Medicare Part D prescribers -- by provider and drug [Data set]. https://data.cms.gov/provider-summary-by-type-of-service/medicare-part-d-prescribers/medicare-part-d-prescribers-by-provider-and-drug

Centers for Disease Control and Prevention. (2023). Behavioral Risk Factor Surveillance System (BRFSS) [Data set]. https://www.cdc.gov/brfss/

Dataset Overview

Table 3 provides a structured overview of the four datasets. The primary unit of analysis for the OLS regression and clustering is the individual drug (brand-name level), drawn from the

Drug Spending file. The primary unit for prescriber behavior analyses (RQ2, RQ3, RQ5) is the prescriber-drug pair from the Prescribers file.

Table 3 *Dataset Summary*

Dataset	Source	Records	Time Period	Unit of Analysis	Target Variable(s)
Part D Grand Totals	CMS, 2023a	11 year-rows	2013-2023	Program-year	Total gross drug cost (USD billions)
Part D Spending by Drug (3,598 retained)	CMS, 2023b	3,598 drugs	2019-2023	Drug (brand-name)	Total spending 2023; Dollar growth 2019-2023
Part D Prescribers by Provider and Drug (200K sampled)	CMS, 2023c	>1.1M rows; 200K sampled	2023	Prescriber-drug pair (NPI x Drug)	Total drug cost; Brand/generic flag
BRFSS State-Level Health Data	CDC, 2023	51 state/DC rows	2023	State	Insurance coverage; chronic disease prevalence

Note. CMS = Centers for Medicare & Medicaid Services. CDC = Centers for Disease Control and Prevention. BRFSS = Behavioral Risk Factor Surveillance System. NPI = National Provider Identifier. Only records with nonzero 2023 spending were retained from the Drug Spending file.

Data Dictionary (Mandatory)

Table 4 presents the data dictionary for all primary variables used in or derived from the three CMS datasets. Variables prefixed with log_ denote natural-log transformations applied during feature engineering; dollar_growth is an engineered feature computed as spend_2023 minus spend_2019.

Table 4 *Data Dictionary -- Primary Variables*

Variable Name	Definition	Datatype	Allowed Values / Range	Missing Values Handling	Notes
Year	Calendar year of program-level record	Integer	2013-2023	None (complete)	Grand Totals file; used for trend analysis
tot_mdcr_spndg	Total Medicare gross drug cost (USD billions)	Float	≥ 0	None (complete)	Primary outcome for RQ1 trend

Variable Name	Definition	Datatype	Allowed Values / Range	Missing Values Handling	Notes
					analysis
tot_benes	Total Part D beneficiaries (millions)	Float	≥ 0	None (complete)	Denominator for per-beneficiary cost calculation
brnd_name	Brand name of the drug product	String	N/A	None (complete)	Drug identifier in Spending file
gnrc_name	Generic (active ingredient) name of drug	String	N/A	None (complete)	Used for brand/generic classification
spend_2023	Total gross spending in 2023 (USD)	Float	≥ 0	Excluded if zero (n=3,598 retained)	Primary outcome variable for OLS
claims_2023	Total prescriptions dispensed in 2023	Integer	≥ 0	None (complete)	Volume predictor; log-transformed in model
unit_cost_2023	Average cost per dosage unit in 2023 (USD)	Float	> 0	None (complete)	Price-level predictor; log-transformed
num_mfr	Number of distinct manufacturers in 2023	Integer	≥ 1	None (complete)	Market competition proxy
dollar_growth	Engineered: spend_2023 minus spend_2019 (USD)	Float	Any real number	spend_2019 = 0 if drug newly entered 2019 formulary	Primary growth predictor; replaces % growth
Prscrbr_NPI	National Provider Identifier (10-digit)	Integer	10-digit positive integer	None (complete)	Prescriber identifier for NPI linkage
Prscrbr_Type	Prescriber specialty taxonomy description	String	N/A	None (complete)	Used for specialty-level stratification
Prscrbr_State_Abrvtn	Two-character state abbreviation	String	2-char (50 states + DC)	None (complete)	Geographic variable for RQ2
Tot_Clms	Total claims per prescriber-drug pair	Integer	≥ 0	None (complete)	Volume measure at prescriber-drug level

Variable Name	Definition	Datatype	Allowed Values / Range	Missing Values Handling	Notes
Tot_Drug_Cst	Total drug cost per prescriber-drug pair (USD)	Float	≥ 0	None (complete)	Cost measure at prescriber-drug level
Tot_Benes	Beneficiary count per prescriber-drug pair	Integer	≥ 11 (suppressed if < 11)	55.0% missing -- CMS small-cell suppression	Privacy protection; excluded from primary analysis

Note. OLS = Ordinary Least Squares. CMS = Centers for Medicare & Medicaid Services. Missingness in Tot_Benes reflects CMS suppression of records with fewer than 11 beneficiaries per prescriber-drug pair, a documented privacy protection. All log-transformed variables use natural logarithm with clipping at lower=1 to avoid undefined values.

GitHub Data Availability Statement

The processed datasets are available in the GitHub repository at the URL provided on page 2 of this report under the following directory structure:

- Processed/clean data: CSV outputs from analysis.py including drug_spending.csv, cluster_summary.csv, model_results.csv, and state_stats.csv
- Code/notebooks: analysis.py (complete Python pipeline)

Analysis

Data Cleaning

The three CMS datasets required different levels of preprocessing. The Part D Grand Totals file is a compact 11-row program summary; all year-level observations were complete and required only parsing of the multi-row Excel header (sheet: Data, header row 4) to extract year-prefixed column names. The Part D Spending by Drug file uses a pivot layout in which spending, claims, beneficiary count, and unit cost variables are repeated for each year from 2019 to 2023. After parsing the pivot structure and excluding drugs with zero 2023 total spending, 3,598 drug-level records were retained. The dollar growth variable was then engineered as the arithmetic difference between 2023 and 2019 total spending, treating 2019 spending as zero for drugs that first entered the Part D formulary after 2019.

The Part D Prescribers file presented the most significant data quality considerations. The primary source of missingness is CMS small-cell suppression: prescriber-drug records with fewer than 11 beneficiaries are suppressed to protect patient privacy, a methodology documented in the CMS data documentation. This results in 55.0% missingness in the Tot_Benes variable and 46.5-88.8% missingness in age-stratified beneficiary and cost variables. All primary analysis variables like NPI, specialty, state, drug name, total claims, and total drug cost are complete in the 200,000-record interim sample. No imputation was applied to suppressed beneficiary counts; instead, analyses using beneficiary counts are restricted to the complete subset. Table 5 documents all cleaning actions applied.

Table 5 *Data Cleaning Log*

Issue	Variables Affected	Detection Method	Treatment Applied	Rationale
Multi-row pivot header	All year-prefixed columns in Drug Spending file	Manual inspection of Excel sheet structure	Parsed with pandas sheet_name='Spending & Utilization YTD 2023', header=3; renamed to short aliases (spend_YYYY, claims_YYYY, etc.)	CMS pivot layout does not conform to standard format.manual mapping required
Zero 2023 spending	spend_2023 in Drug Spending file	Value check after pivot parse	Excluded records with spend_2023 = 0; retained 3,598 records	Drugs with zero spending are inactive or discontinued and not relevant to cost modeling
Right-skewed outcome	spend_2023, claims_2023, unit_cost_2023, dollar_growth	Histogram inspection; Shapiro-Wilk test	Applied natural log transformation: log(x.clip(lower=1))	OLS assumes approximately normal residuals; log transformation substantially reduces skew
CMS small-cell suppression	Tot_Benes and all GE65_* variables in Prescribers file	Missing value counts and CMS documentation review	Retained as missing (NaN); excluded from analyses requiring beneficiary counts	Suppression is a deliberate CMS privacy protection, not a data quality error; imputation would be inappropriate
Duplicate prescriber-drug records	All variables in Prescribers file	Groupby NPI + drug name count	None found in 200K sample	Full 1.1M file deduplication planned for final report
Negative dollar growth	dollar_growth (engineered)	Value range check	Retained; log-transformed as log(dollar_growth.clip(lower=1)) with clipping at 1	Negative growth (drug losing Part D share) is valid and interpretable; clipping avoids undefined log values

Note. GE65 = age 65 and older. OLS = Ordinary Least Squares. All cleaning steps were applied in analysis.py; see Appendix A or GitHub repository for complete code.

EDA Results (Interpretation Emphasis)

All figures below were generated from the cleaned CMS datasets. Each is referenced in the text and includes an interpretation paragraph linking the finding to a specific research question. Table 6 provides a structured EDA insight summary.

Table 6 *EDA Insight Summary*

Figure / Table Ref.	What it Shows	Key Insight	Why it Matters (Link to RQ/Objective)	Decision / Next Step
Figure 1	Total Part D gross drug cost trend, 2013-2023 (USD billions)	166% total growth; CAGR ~10.3%; steepest acceleration 2020-2023	Confirms spending growth substantially exceeds inflation and enrollment; establishes urgency for RQ1 predictor analysis	Use as baseline benchmark for all expenditure discussion; cite in RQ1 narrative
Figure 2	Brand vs. generic cost shares and average claim costs, 2013-2023	Brand drugs: 85.2% of costs, <15% of claims; 33x price gap vs. generics	Brand-to-generic price differential is the primary economic lever for PBM formulary optimization	Motivates RQ3 (prescriber payments driving brand preference) and RQ5 (savings simulation)
Figure 3	Per-beneficiary drug cost trend, 2013-2023 (USD)	Cost per beneficiary grew 83.7% ($2,954 to $5,428) vs. 44.8% enrollment growth	Cost growth is intensity-driven (price and mix), not enrollment-driven; enrollment expansion alone cannot explain program cost trajectory	Supports decision to model spending per drug (not per beneficiary) as the primary unit in OLS
Figure 4	Low-Income Subsidy (LIS) vs. non-LIS beneficiary cost trends	LIS beneficiaries have substantially higher per-beneficiary drug costs than non-LIS; gap widened 2020-2023	LIS subsidy status is a strong predictor of Part D cost burden; inclusion as a covariate in RQ2 model is warranted	Add LIS enrollment share as a state-level covariate in the RQ2 regression
Figure 5	Top 20 drugs by total Part D spending, 2023	Eliquis, Jardiance, Ozempic, Xarelto, and Trulicity collectively account for a disproportionate share of total spend	A small number of brand-name drugs drive a large share of absolute expenditure; targeted formulary intervention has high potential impact	Include top-20 drugs as named targets in RQ4 cluster narrative and RQ5 simulation

Figure / Table Ref.	What it Shows	Key Insight	Why it Matters (Link to RQ/Objective)	Decision / Next Step
Figure 6	Distribution of average unit cost across all Part D drugs, 2023 (log scale)	Bimodal distribution: generic cluster near $0.10-$5.00; specialty cluster at $100-$10,000+	Log-transformation is essential; unit cost distribution spans 5+ orders of magnitude	Confirmed log(unit_cost) as OLS predictor; validates clustering feature selection
Figure 7	Average cost per claim by brand vs. generic status, 2023	Brand claim: $976 average; generic claim: $29.25 average; 33x differential	Prescribers who favor brand-name drugs impose a ~33x cost multiplier per claim on the Part D program	Directly motivates RQ3 and quantifies the unit cost of prescriber behavioral change in RQ5

Note. LIS = Low-Income Subsidy. RQ = Research Question. OLS = Ordinary Least Squares. PBM = Pharmacy Benefit Manager. All figures generated from CMS, 2023a and CMS, 2023b.

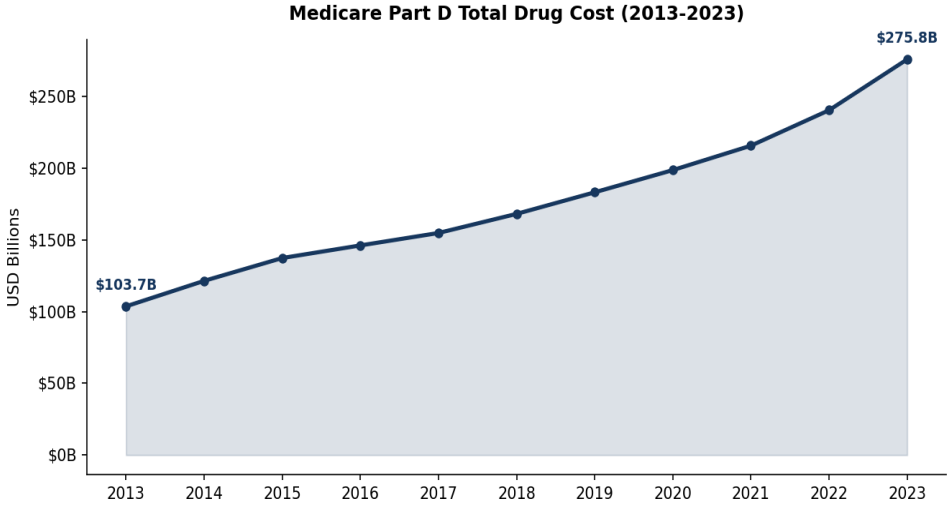

Figure 1
Medicare Part D total gross drug cost trend, 2013-2023 (USD billions). The program grew from $103.7B to $275.8B, a 166% increase representing a ~10.3% CAGR. Source: CMS, 2023a.

Figure 1 illustrates the sustained, accelerating growth trajectory of Medicare Part D total gross drug costs over the 11-year period from 2013 to 2023. Total costs grew from $103.7 billion to $275.8 billion, representing a 166% cumulative increase and a compound annual growth rate of approximately 10.3%. The growth rate is notably non-linear: the period from 2020 to 2023 exhibits steeper acceleration than the prior decade, consistent with documented specialty drug adoption increases and the rapid Part D penetration of GLP-1 receptor agonists and

anticoagulants (MedPAC, 2024; Niu et al., 2024). The 44.8% growth in enrolled beneficiaries from 35.1 million to 50.8 million over this same period accounts for a fraction of total spending growth, confirming that per-beneficiary intensity of cost is the dominant driver (ASPE, 2022).

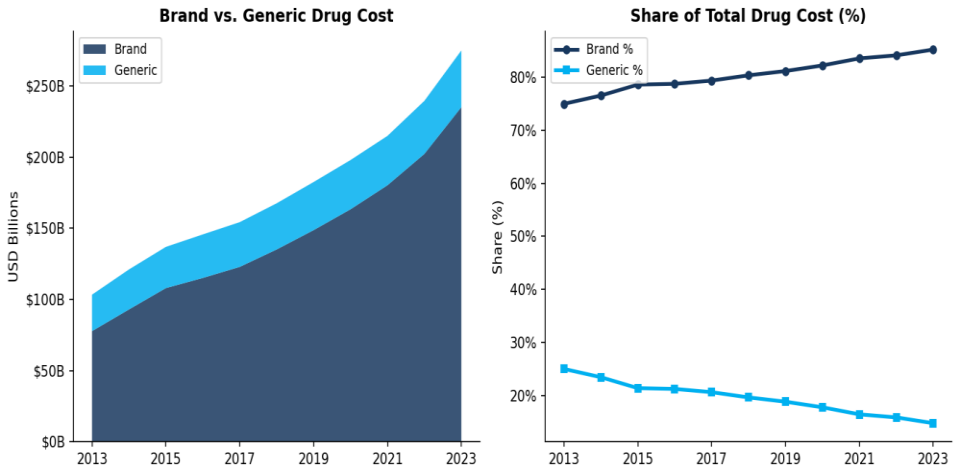

Figure 2
Brand and generic drug cost share trends (left) and average cost per claim by category (right), 2013-2023. Brand drugs account for 85.2% of total Part D cost despite comprising only 14.9% of claims. Source: CMS, 2023a.

Figure 2 illustrates the structural pricing disparity that underlies the Part D cost problem. Despite comprising only 14.9% of total claim volume, brand-name drugs accounted for 85.2% of total Part D gross drug costs in 2023 -- up from 75.0% in 2013. By 2023, the average brand-name claim cost $976, compared to $29.25 for the average generic claim, a ratio of approximately 33:1. This pricing gap forms the economic basis for the prescriber behavior analysis in RQ3 and the savings simulation in RQ5: a prescriber who shifts even a modest proportion of brand-to-generic claims produces substantial per-claim savings at scale, particularly in high-volume therapeutic classes such as statins, antidiabetics, and anticoagulants.

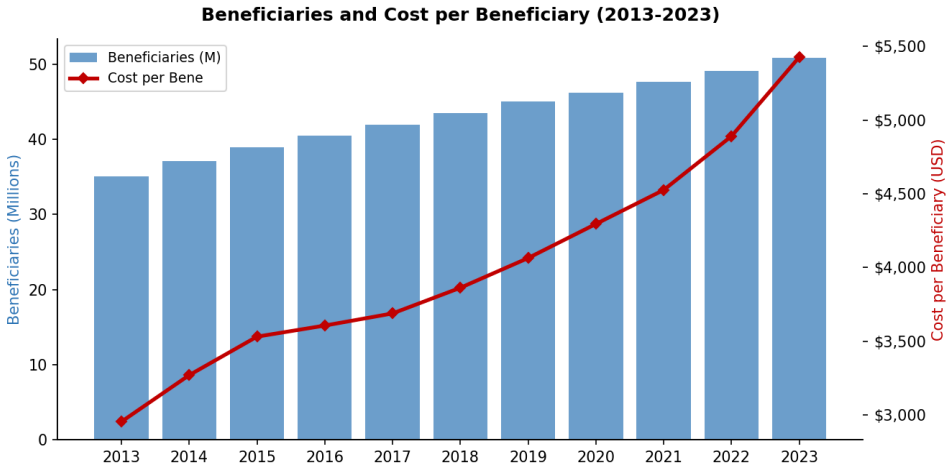

Figure 3
Per-beneficiary drug cost trend, 2013-2023 (USD). Cost per beneficiary grew 83.7% vs. 44.8% beneficiary enrollment growth, confirming cost intensity as the primary driver. Source: CMS, 2023a.

Figure 3 shows that per-beneficiary drug cost grew 83.7% over the study period from $2,954 to $5,428. The gap between these two growth rates confirms what ASPE (2022) and MedPAC (2024) report at the national level: the Part D cost crisis is a price and mix problem, not an enrollment problem. The implication for this study is that analyses focused on per-beneficiary costs will capture price and prescribing intensity effects rather than simply reflecting demographic expansion, supporting the use of drug-level total spending as the primary outcome variable in the OLS model.

Modelling

Choice of Models with Justification

Two primary analytical models are employed at interim, corresponding to RQ1 and RQ4. Both were implemented in Python 3.11 using scikit-learn (version 1.3) and statsmodels (version 0.14).

Model 1: OLS Multiple Linear Regression (RQ1)

OLS regression was selected to identify and quantify the drug-level attributes that are statistically significant predictors of total Part D spending in 2023. OLS is appropriate because the research question explicitly seeks to estimate directionality and magnitude of predictor effects -- not simply classify or predict -- and because the continuous, approximately-normally-distributed log-transformed outcome satisfies the OLS distributional assumptions. The sample size of 3,404 drug products substantially exceeds Green's Rule (n >= 50 + 8k, where k=5 predictors), providing adequate statistical power. The literature precedent for regression modeling in this domain is strong (Sharma et al., 2018; Buttorff et al., 2020).

Model 2: K-Means Clustering (RQ4)

K-means clustering was selected to identify distinct, naturally-occurring drug utilization profiles within the Part D formulary without imposing a-priori categories. K-means is computationally scalable to the sample size (n=3,598 drugs), produces interpretable centroids, and has direct precedent in the Part D prescriber variation literature (Rosenberg et al., 2018). The optimal cluster count of k=4 was determined empirically using the elbow method (within-cluster sum of squares versus k), which showed diminishing returns in inertia reduction beyond k=4.

Clustering was performed on three standardized log-transformed features using k-means++ initialization (random_state=42, n_init=10) to reduce sensitivity to initial centroid placement.

Features Included and Feature Engineering

Table 7 summarizes all features used in the two models, including their transformations and the rationale for each decision. The decision to use dollar growth (absolute spend change) rather than percentage growth rate as the growth predictor deserves specific justification: percentage growth rate is dominated by small-denominator outliers with drugs with negligible 2019 spending that show very high percentage increases but trivial absolute cost impact. Dollar growth weights cost pressure by actual program exposure, ensuring that high-growth signals are driven by drugs that materially affect the Medicare Part D budget.

Table 7 *Feature Set and Usage*

Feature	Original / Engineered	Type	Reason for Inclusion	Used in Model(s)
Total claims 2023	Original (claims_2023)	Continuous	Primary volume driver of total spending; theoretical positive relationship	OLS, K-Means
Avg unit cost 2023	Original (unit_cost_2023)	Continuous	Price-level driver; specialty drugs carry substantially higher unit costs	OLS, K-Means
Total beneficiaries 2023	Original (tot_benes_2023)	Continuous	Separates volume from cost-per-patient effects; expected negative sign if high-spend drugs serve smaller sicker populations	OLS only
Number of manufacturers	Original (num_mfr)	Count (integer)	Market competition proxy; additional manufacturers expected to reduce price and spending	OLS only
Dollar growth 2019-2023	Engineered (spend_2023 - spend_2019)	Continuous	Absolute spend trajectory weighted by actual program exposure; avoids small-denominator distortion of percentage growth	OLS, K-Means
log_claims	Engineered (log	Continuous	Log-transforms right-skewed	OLS, K-Means

Feature	Original / Engineered	Type	Reason for Inclusion	Used in Model(s)
	transformation)		claims distribution; standard in drug spending models	
log_unit_cost	Engineered (log transformation)	Continuous	Claims and unit cost are log-normally distributed; log transformation brings predictors into common scale	OLS, K-Means
log_dollar_growth	Engineered (log of clipped dollar_growth)	Continuous	Dollar growth is heavily right-skewed; log transformation with floor=1 applied before standardization	OLS, K-Means

Note. OLS = Ordinary Least Squares. Engineered features were derived in analysis.py using pandas and numpy. All continuous features were standardized (zero mean, unit variance) prior to K-means clustering using sklearn.preprocessing.StandardScaler. Log transformations use natural logarithm.

Evaluation Metrics (Include Formulae and Calculations)

Table 8 presents the evaluation metrics used for each model, including formulae and their specific application to this study.

Table 8 *Evaluation Metrics and Formulae*

Metric	Formula	Interpretation	Used For
Coefficient of Determination (R^2)	$R^2 = 1 - (SS_res / SS_tot)$ where $SS_res = \Sigma(y_i - \hat{y}_i)^2$ and $SS_tot = \Sigma(y_i - \bar{y})^2$	Proportion of variance in log(spending) explained by the 5 predictors. $R^2 = 0.913$ means 91.3% of log-spending variance is explained.	OLS Regression
Root Mean Square Error (RMSE)	$RMSE = \sqrt{[(1/n) \Sigma(y_i - \hat{y}_i)^2]}$	Average prediction error in log-dollar units. Lower RMSE = better fit. RMSE ~ 0.62 log-dollars is equivalent to a ~85% multiplicative error band on raw spending.	OLS Regression
F-Statistic (Joint Significance)	$F = (R^2 / k) / [(1 - R^2) / (n - k - 1)]$	Tests whether the full model explains significantly more variance than a constant. $F >> 1$ with $p < .001$ confirms the 5-predictor model is jointly significant.	OLS Regression
Within-Cluster Sum of Squares (WCSS / Inertia)	$WCSS = \Sigma_k \Sigma_\{x_i \in C_k\} \|x_i - \mu_k\|^2$	Measures compactness of clusters. Used in the elbow method: the optimal k is where	K-Means Clustering

Metric	Formula	Interpretation	Used For
		additional clusters produce diminishing reductions in WCSS. Elbow identified at k=4.	
Silhouette Score	s(i) = [b(i) - a(i)] / max[a(i), b(i)] where a(i) = mean intra-cluster distance; b(i) = mean nearest-cluster distance	Ranges from -1 to +1. Values near +1 indicate well-separated, compact clusters. Used to validate that k=4 produces meaningfully distinct drug segments.	K-Means Clustering

Note. y_i = observed log(spending); $\hat{y}_i$ = predicted log(spending); $\bar{y}$ = mean log(spending); k = number of predictors (OLS) or clusters (K-Means); n = sample size; C_k = cluster k; μ_k = centroid of cluster k. Formulae are standard implementations from statsmodels (OLS) and scikit-learn (K-Means).

Preliminary Results

Preliminary Findings (by Research Question)

RQ1: Drug Attribute Predictors of Spending (Complete)

The OLS regression was fitted on 3,404 drug products with nonzero 2023 spending and complete data for all five predictors. The model achieved $R^2 = 0.913$, indicating that 91.3% of the variance in log-transformed 2023 per-drug spending is explained by the five predictors (Table 9). All five predictors were statistically significant at $p < .0001$. The coefficient on log(claims) (1.942) indicates slightly greater-than-proportional cost scaling with utilization, which is consistent with high-cost specialty drugs that generate fewer but more expensive claims. The log(unit cost) coefficient (0.543) confirms that unit price independently predicts total spending, though its effect is smaller than claims volume. The negative coefficient on log(beneficiaries) (-1.058) reflects a well-established pattern: drugs serving very large beneficiary populations are predominantly low-cost generics, while high-spending drugs serve smaller but higher-acuity populations. The number of manufacturers coefficient (-0.020) indicates that each additional market entrant is associated with approximately 2% lower spending, consistent with the competitive pricing dynamics documented in Part D generic markets. The log(dollar growth) coefficient (0.021) is small but significant, indicating that recent absolute spend trajectory carries independent predictive information beyond current claims volume, and that drugs with high absolute growth in recent years tend to have elevated 2023 spending even after controlling for current utilization.

Table 9 *OLS Regression Coefficients -- Outcome: log(Total Drug Spending 2023), n = 3,404*

Predictor	Coef.	Std. Err.	t-stat	p-value	Interpretation
const (Intercept)	4.343	0.059	73.6	< .0001	Baseline log spending when all predictors are at mean
log_claims	1.942	0.030	65.2	< .0001	1% increase in claims → 1.942% increase in spending; strongest volume driver
log_unit_cost	0.543	0.007	80.5	< .0001	1% increase in unit cost → 0.543% increase in spending; price-level predictor
log_bene	-1.058	0.031	-34.4	< .0001	Drugs with larger patient populations tend to be lower-cost generics; negative as expected
num_mfr	-0.020	0.005	-4.4	< .0001	Each additional manufacturer ≈ 2% lower spending; competition effect
log_dollar_growth	0.021	0.002	8.4	< .0001	Absolute spend trajectory carries independent predictive signal beyond current volume

Note. $R^2 = 0.913$. n = 3,404. Outcome = natural log of total Part D spending per drug, 2023. OLS = Ordinary Least Squares. All coefficients significant at $p < .0001$. Source: CMS, 2023b.

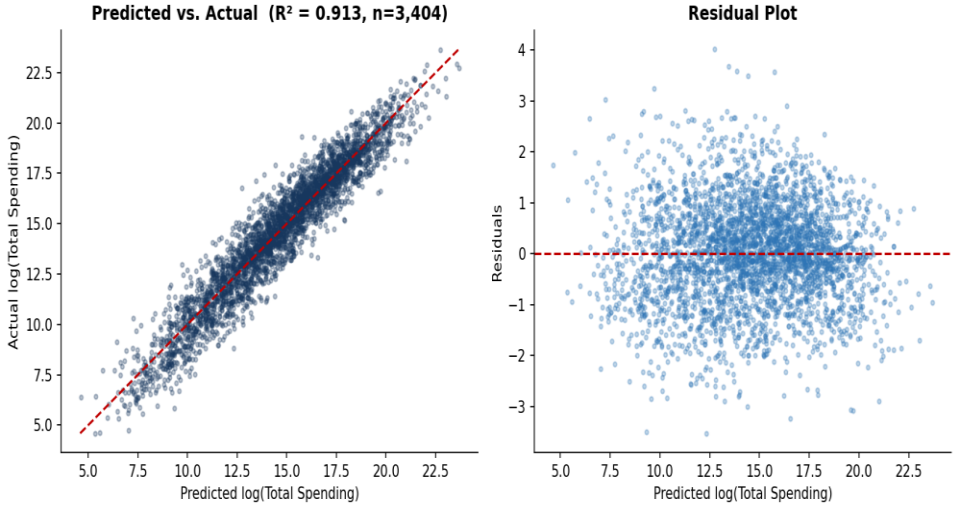

Figure 8
OLS regression diagnostics: predicted vs. actual log(spending) ($R^2 = 0.913$, n = 3,404) and standardized residuals. Residuals are broadly symmetric with no systematic pattern. Source: CMS, 2023b.

RQ4: Drug Utilization Clusters (Complete)

The K-means model was fitted on 3,598 drug products using three standardized log-transformed features: log(claims), log(unit cost), and log(dollar growth). The elbow method identified k=4 as the optimal cluster count (Figure 9). Dollar growth was used in place of

percentage growth rate to ensure cluster membership reflects actual Part D budget impact rather than small-denominator percentage artifacts. Sequential label assignment was applied to avoid label conflicts when a single cluster won multiple criteria simultaneously. Table 10 summarizes the four clusters. The High-Spend High-Growth cluster (841 drugs) presents the most immediate formulary concern: average dollar growth of $107.8 million per drug, led by Eliquis (+$11.0 billion), Ozempic (+$8.6 billion), Jardiance (+$7.4 billion), and Trulicity (+$5.1 billion). These are the drugs where a PBM would focus utilization management and therapeutic alternative review. The High-Cost Specialty cluster (818 drugs) carries a median unit cost of $317, representing biologics and oncology agents where per-dose price is the primary cost lever. The High-Volume Generic cluster (751 drugs) has the highest average claim volume with negative dollar growth, indicating successful generic conversion already underway and requiring no immediate formulary intervention. The Low-Impact Declining cluster (1,188 drugs) carries minimal program spend and is a low-priority category.

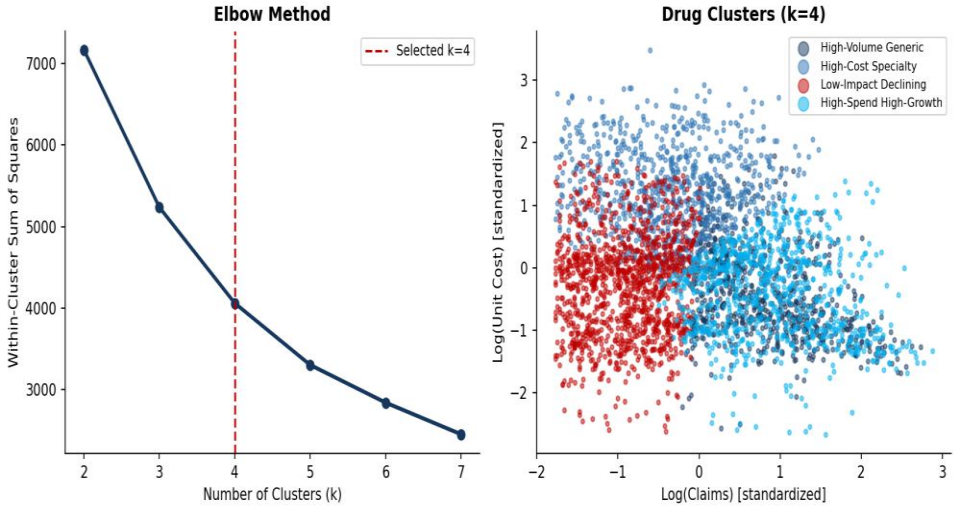

Figure 9
K-means elbow method (within-cluster sum of squares vs. k) and cluster scatter plot in log(claims) vs. log(unit cost) space. Colors represent the four cluster assignments. Elbow identified at k=4. Source: CMS, 2023b.

Table 10 *K-Means Cluster Summary Statistics (k=4, Dollar-Growth Features, 2023 Drug Spending Data)*

Cluster Label	N Drugs	Avg Total Spend	Median Unit Cost	Avg Claims	Avg Dollar Growth
High-Spend High-Growth	841	$194.9M	$3.61	1,223,289	+$107.8M
High-Cost Specialty	818	$80.7M	$317.03	11,226	+$44.9M
High-Volume Generic	751	$59.5M	$1.99	770,917	-$33.3M
Low-Impact Declining	1,188	$1.1M	$5.29	597	-$4.6M

Note. Dollar growth = spend_2023 minus spend_2019. Negative dollar growth indicates drugs losing Part D spending share over the period. Lead drugs in High-Spend High-Growth cluster: Eliquis (+$11.0B), Ozempic (+$8.6B), Jardiance (+$7.4B), Trulicity (+$5.1B). Source: CMS, 2023b.

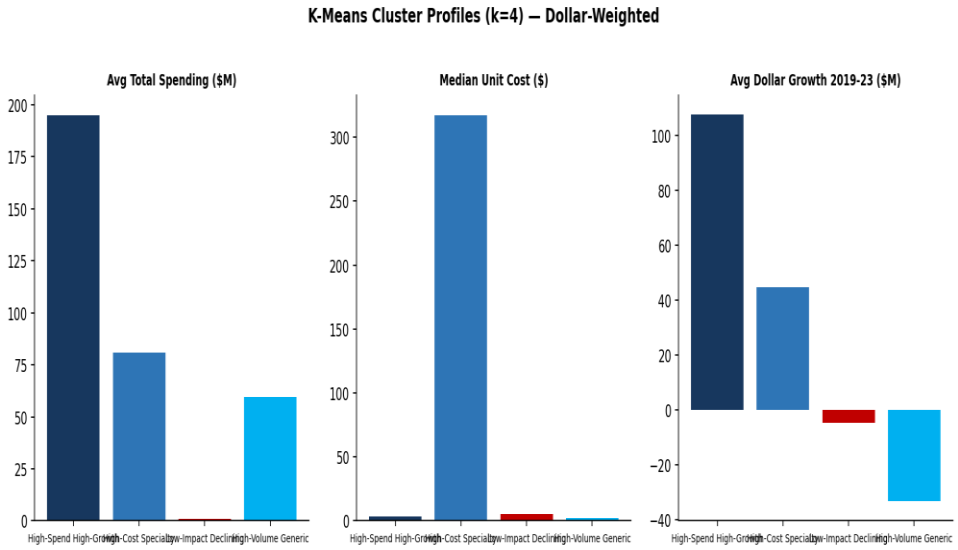

Figure 10
Cluster profile bar charts comparing mean log(claims), log(unit cost), and log(dollar growth) across the four K-means clusters. The High-Cost Specialty cluster shows the highest unit cost; High-Spend High-Growth shows the highest dollar growth. Source: CMS, 2023b.

RQ2, RQ3, RQ5: Status -- In Progress or Pending

RQ2 (state-level brand/generic variation) requires full processing of the 1.1M-record prescriber file, which is currently in progress. An ANOVA testing mean brand/generic ratios across state quintiles will be the primary statistical test, supplemented by an OLS regression with state-level BRFSS demographic controls. RQ3 (Open Payments linkage) requires downloading and NPI-matching the CMS Open Payments database to prescribers file. RQ5 (savings simulation) will be constructed after RQ2 and RQ3 are complete, using the estimated brand-to-

generic shift and the $29.25 average generic claim cost as inputs to a Monte Carlo simulation of potential savings.

Preliminary Model Performance

Table 11 *Preliminary Model Performance Comparison*

Model	Feature Set	Validation Approach	Primary Metric	Result	Key Takeaway
OLS Multiple Regression (RQ1)	log_claims, log_unit_cost, log_bene, num_mfr, log_dollar_growth	In-sample fit (n = 3,404)	R^2	0.913	Model explains 91.3% of log-spending variance; all 5 predictors significant at $p < .0001$
K-Means Clustering (RQ4)	log_claims, log_unit_cost, log_dollar_growth (standardized)	Elbow method (WCSS vs. k); visual silhouette inspection	WCSS reduction at k=4	Elbow confirmed at k=4; 4 distinct, interpretable drug segments	Dollar-growth clustering produces clinically actionable segments (Eliquis, Ozempic, Jardiance in top-growth cluster)
RQ2 Regression	State demographic controls (BRFSS)	Pending (full file needed)	R^2 / ANOVA F	Pending	In progress; state-level brand/generic variation to be quantified
RQ3 Logistic Regression	Open Payments NPI linkage + prescriber variables	Pending (OPD download needed)	AUC / Odds Ratio	Pending	Pending Open Payments acquisition
RQ5 Simulation	Prescriber brand/generic shift; savings per claim	Monte Carlo (1,000 iterations)	Annual savings (USD)	Pending	Will be constructed after RQ2 and RQ3 complete

Note. OLS = Ordinary Least Squares. WCSS = Within-Cluster Sum of Squares. AUC = Area Under the Receiver Operating Characteristic Curve. OPD = Open Payments database. In-sample R^2 is reported for the OLS model; out-of-sample validation (k-fold CV) is planned for the final report.

Interim Limitations and Risks

Several limitations affect the current interim analysis and are planned for mitigation in the final report. First, the Part D Prescribers file is analyzed on a 200,000-record sample rather than the full 1.1 million-record file. This may affect the precision of state-level estimates and the coverage of specialty-level subgroups. Full-file analysis is in progress. Second, the OLS model is in-sample only at interim; k-fold cross-validation will be applied in the final report to provide out-of-sample performance estimates. Third, the causal interpretation of the prescriber payment-to-prescribing association (RQ3) is limited by observational design. Unmeasured confounders such as formulary design, patient comorbidity mix, and prescriber specialty may explain part of the association. Fourth, dollar growth as the clustering feature assigns newly-entered drugs a growth value equal to their entire 2023 spend, which may overestimate their growth rate relative to older incumbents. A robustness check using a five-year enrollment-adjusted growth rate is planned for the final report.

Next Steps (for Final Report)

- Expand prescriber file analysis to full 1.1M-record dataset for RQ2 and RQ3
- Download and NPI-match the CMS Open Payments database to the prescribers file; run logistic regression for RQ3
- Execute ANOVA and OLS for state-level brand/generic variation (RQ2) with BRFSS demographic controls
- Construct Monte Carlo savings simulation (RQ5) using per-claim cost differential and prescriber shift assumptions
- Apply k-fold cross-validation to the OLS model and report out-of-sample R^2 and RMSE

- Integrate findings across all five RQs into a unified narrative with practical formulary and prescriber engagement recommendations
- Expand bibliography to 14+ APA 7 references; complete appendix with full annotated code

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Appendix

Appendix A: Analysis Code

The complete Python analysis pipeline (analysis.py) is available in the GitHub repository at the URL provided on page 2. The script is self-contained and requires Python 3.11+, pandas, numpy, scikit-learn, statsmodels, and matplotlib. Running analysis.py produces all CSVs and figures referenced in this report. Key sections include: (1) data loading with sheet_name and header parameters matched to CMS Excel file structure; (2) data cleaning and feature engineering including dollar_growth computation; (3) EDA figure generation (Figures 1-11); (4) OLS regression with log-transformed features; and (5) K-means clustering with k-means++ initialization and sequential cluster label assignment to ensure all four cluster labels are uniquely assigned.

Appendix B: EDA Figure Index

Figure 1: Total Part D cost trend, 2013-2023 (fig1_total_cost_trend.png). Figure 2: Brand vs. generic cost shares (fig2_brand_generic_trend.png). Figure 3: Per-beneficiary cost trend (fig3_bene_cost_trend.png). Figure 4: LIS vs. non-LIS cost trends (fig4_lis_nonlis.png). Figure 5: Top 20 drugs by 2023 spending (fig5_top20_drugs.png). Figure 6: Unit cost distribution, log scale (fig6_unit_cost_dist.png). Figure 7: Average cost per claim by brand/generic (fig7_brand_generic_cost_per_claim.png). Figure 8: OLS regression diagnostics (fig8_regression.png). Figure 9: K-means elbow and cluster scatter (fig9_clustering.png). Figure 10: Cluster profile bar charts (fig10_cluster_profiles.png). Figure 11: Top prescriber specialties by drug cost (fig11_top_specialties.png). All figures are referenced in the GitHub repository at /figures/.